LAPORAN TUGAS AKHIR
HUMAN-COMPUTER
INTERACTION & UI/UX

Redesigning the K24Klik Application Using a Design Thinking Approach

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CHAPTER 1: INTRODUCTION

1.1 Background & App Selection

In today's digital era, health and pharmacy applications are essential. However, not all applications provide an optimal User Experience (UX). For this project, the **K24Klik** application was selected, specifically focusing on its usage in local branches (K-24 AW Sumarmo Purbalingga & Dukuhwaluh, Banyumas). Although K24Klik is a national platform, its current User Interface (UI) suffers from severe visual clutter, invasive pop-ups, and cognitive overload, which hinders local users from quickly ordering medicine during emergencies.

1.2 App Info

[View App on Google Drive]

1.2.1 App Information & Play Store Listing

App Identity & Metadata:

- **App Name:** K24KLIK: Beli Obat Cepat 24mnt
- **Developer:** PT. K24 Klik Indonesia
- **App Category:** Medical / Health & Fitness (Kesehatan)
- **Current Version:** v5.215.14 *(Evidenced in screenshot #18/64)*
- **Last Updated:** April 15, 2026 *(Evidenced in screenshot #159)*
- **Downloads:** 1,000,000+ (1M+)
- **Play Store Rating:** 4.2 / 5.0
- **Total Reviews:** 44K+ reviews
- **Core Tagline:** *"Pesan obat dari rumah lebih cepat loh, 24 menit sudah sampai!"*
- **Target Users (Demographic):** General public, patients needing urgent medication, mothers buying baby supplies (susu/vitamin), and elderly patients (or their caretakers) needing routine prescriptions.

Core Services / Features:

1. **Medicine E-Commerce:** Over-the-counter (OTC) and prescription drugs.
2. **Delivery Options:** 24-Minute Delivery (Prioritas), 1-Hour Delivery (Instan), Regular, and "Ambil di Apotek" (Pick-up).
3. **Telemedicine:** "Dokter Siaga Online" (Konsultasi Rp 15.000).
4. **Prescription Upload:** "Unggah Resep" with digital and physical upload options.
5. **Loyalty Program:** K24Klik Points & Vouchers.

1.2.2 UI/UX, HCI, & Core Design Identity

Visual Identity (Brand Guidelines):

- **Primary Color:** K-24 Green (Hex: #2E7D32 or similar emerald green).
- **Secondary/Accent Colors:** Yellow/Orange (used for warnings, points, and promo badges).
- **Background:** White / Off-white (Hex: #FFFFFF / #F5F5F5).
- **Typography:** Sans-serif (Clean, modern, readable).

HCI & Ergonomic Evaluation of Current App (Based on slides):

- **Gestalt Principle Violation (Proximity & Figure-Ground):** The Home Screen is extremely cluttered. Banners, feature icons (Konsultasi, Unggah Resep, Vaksinasi), points, and categories are all crammed together. The user's eye doesn't know where to focus.
- **Ergonomics (Mental Workload):** The cognitive load to buy a simple medicine is too high. Users are bombarded with pop-ups ("Enable Location", "Dapatkan Cashback", "Ketentuan Pembelian") before they can even act.
- **Thumb Zone (Mobile Usability):** While the bottom navigation bar is good, many crucial filters and the "X" buttons to close pop-ups are at the very top of the screen, making one-handed use difficult.

1.2.3 App Screens & User Flow (The Current Architecture)

Here is the current, highly complex flow of the app (which was simplified into just 4 screens for the MVP):

1. **Onboarding/Splash:** 5 swipeable screens explaining 24-min delivery, nearest pharmacy selection, and cart mechanics.
2. **Login/Register:** Offers Google Sign-In or manual Email/Phone + Password.
3. **Home Screen:**
    - Location selector at the top.
    - 4 Main Feature Buttons (Konsultasi, Resep, Vaksin, Dokter).
    - Promo Banners (Claim Vouchermu).
    - Categories Grid (Promo, Susu, Vitamin, Kontrasepsi).
    - Horizontal scrolling lists for "Terlaris" (Best Sellers).
4. **Search & Filter:** *Live search with auto-suggest and chips for tags like alergi, gatal, generik.*
5. **Product Detail Page (PDP):**
    - Product Image & Price.
    - *Massive blocks of text for Komposisi, Indikasi, Dosis, Cara Penyimpanan, Efek Samping.* (This requires endless scrolling).

Tanya Jawab Apoteker (Q&A section).

6. **Cart & Checkout:**

    Cart summary with +/- quantity buttons.

    Popup warning for minimum purchase (Rp 10.000).

    Select Delivery Method & Address.

    Select Payment Method (E-Wallet, Transfer BCA/Mandiri/BNI).

1.2.4 Representative User Reviews (Real Data Evidence)

For the report, real user feedback was analyzed. These categorized reviews serve as "App Store Review Analysis" data to justify the redesign.

Positive Reviews (What works):

- *(Budi S.) - "Sangat membantu saat malam hari butuh obat demam untuk anak. Pengiriman benar-benar cepat 24 menit dari apotek K-24 Dukuhwaluh."* (Insight: Local branch fulfillment and speed are the core values).
- *(Rina M.) - "Obatnya komplit dan pasti asli. Lumayan sering dapat promo gratis ongkir."*

Negative / Critical Reviews (The UX Pain Points we will fix):

- *(Ahmad T.) - "Terlalu banyak popup! Baru buka aplikasi langsung disuruh nyalakan lokasi, lalu ada popup banner promo. Bikin pusing, padahal saya cuma mau cepat cari Paracetamol."* (UX Flaw: Violates User Control & Freedom. Fix: Remove invasive popups).
- *(Siti A.) - "Halaman deskripsi obat kepanjangan. Tulisannya kecil-kecil dan numpuk jadi satu. Susah baca efek sampingnya di HP."* (UX Flaw: Violates Aesthetic & Minimalist Design. Fix: Use clean, collapsible Accordion tabs for descriptions).
- *(Dimas P.) - "Checkoutnya ribet. Pas mau bayar disuruh milih cabang apotek sendiri, terus peringatan minimal belanja Rp10.000 munculnya pas mau klik bayar, bukan dari awal."* (UX Flaw: Violates Visibility of System Status & Error Prevention. Fix: Streamline checkout into one clean page, warn users early).

1.3 Challenge Statement

"To redesign the K24Klik application interface so that local users in Purbalingga/Banyumas can search, view details, and order medicine independently 50% faster through a clean, minimalist design free from disruptive pop-ups."

1.4 Methodology Selection

This project utilizes the **Design Thinking** methodology rather than traditional UCD (User-Centered Design). This choice was made because the problem required exploring latent user needs through deep *Empathy* and *Ideation*, allowing for rapid *Prototyping* (based on the Stanford d.school model) to discover the most ergonomic interface solutions.

CHAPTER 2: LITERATURE REVIEW

2.1 HCI and Design Thinking Concepts

Human-Computer Interaction (HCI) is the study of how humans interact with digital technology to create intuitive systems. *Design Thinking* is a human-centered problem-solving method consisting of 5 stages: Empathize, Define, Ideate, Prototype, and Test.

2.2 Heuristics & Design Principles

In evaluating and designing the interface, this report is guided by several core HCI theories:

1. **Jakob Nielsen's 10 Usability Heuristics:** Standard UI evaluation to prevent errors and ensure User Control & Freedom.
2. **Ben Shneiderman's 8 Golden Rules:** Principles for interactive design, such as striving for consistency and offering informative feedback.
3. **CRAP Principles (Robin Patricia Williams):** Contrast, Repetition, Alignment, and Proximity.
4. **Gestalt Principles:** Human visual perception psychology (e.g., *Similarity* and *Figure-Ground*).
5. **Physical & Cognitive Ergonomics:** The application of the "Thumb Zone" and minimum touch target boundaries (Apple HIG: 44x44 pt) for mobile usability.

2.3 Development Technology (Flutter)

The final MVP (Minimum Viable Product) prototype was implemented using **Flutter**, a UI toolkit from Google for building natively compiled, cross-platform applications from a single codebase.

CHAPTER 3: DATA GATHERING

3.1 Heuristic Evaluation of the Current App (Old App)

Before redesigning, the K24Klik application (v5.215.14) was evaluated using Jakob Nielsen's principles.

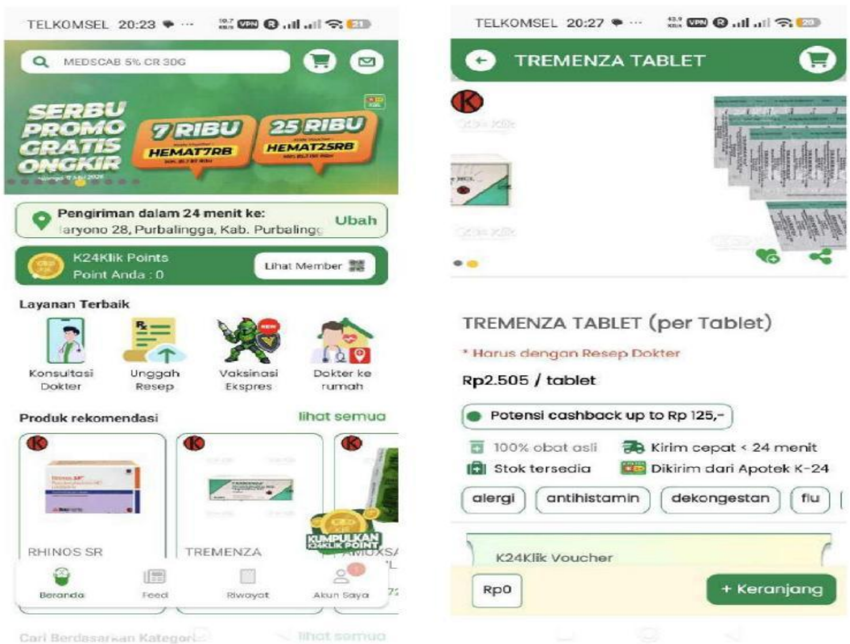

Figure 3.1 – Screenshots of the original K24Klik app (Home Screen & Product Detail Page)

Table 3.1: Heuristic Evaluation of K24Klik (Original)

Nielsen's Heuristic	Observation / Pain Point	Severity (0–4)
8. Aesthetic & Minimalist Design	Visual Clutter: The home screen is overwhelmed with banners, promos, and menus, causing high cognitive load.	**4**
3. User Control & Freedom	Aggressive Pop-ups: Users are blocked by sudden pop-ups (location permissions, promos) before they can even search.	**4**

7. Flexibility & Efficiency	Information Overload: Medicine details are long, stacked blocks of text, requiring endless scrolling.	**3**
1. Visibility of System Status	Confusing Checkout: Minimum purchase warnings only appear as errors at the final payment stage, not at the beginning.	**3**

3.2 Quantitative User Survey

A Google Form survey was distributed, gathering **32 real responses** (majority aged 18–25).

Figure 3.2 – Survey QR Code

Form Link:

https://forms.cloud.microsoft/r/nX6umJxAYV

Key Data Analysis:

6. **UI Preference:** 97% of respondents (31 out of 32) preferred a "Minimalist (clean, neat, simple)" UI over a dense, promo-heavy UI.
7. **Text Readability:** 38% complained that the medicine description text was too long and dizzying to read.
8. **Language Preference:** 66% wanted a bilingual application (Indonesian and English toggle).

Responses Summary Link: View Analysis

Raw Responses Excel File Link: View Data

3.3 Stakeholder Interview Synthesis (Qualitative)

Extracted Interview Script (with local K-24 pharmacy worker):

Q1: "Permisi Kak/Pak/Bu, saya mahasiswa lagi ngerjain tugas evaluasi UI/UX aplikasi K24Klik. Saya survei 32 orang anak muda (18-25 tahun). Ternyata 97% dari mereka bilang tampilan aplikasi K24Klik sekarang terlalu ramai dan mereka pengen yang bersih/minimalis aja. Menurut Kakak yang jaga di toko, apakah pelanggan K-24 memang lebih suka yang simpel-simpel aja kalau beli obat?"

Jawaban: "Iya bener Mas/Mbak. Karakteristik orang beli obat itu beda sama orang belanja baju di Shopee. Orang beli obat itu biasanya lagi sakit, panik, atau buru-buru pengen cepet sembuh. Jadi mereka nggak mau lihat-lihat yang lain. Mendingan aplikasinya langsung to-the-point aja ke kolom pencarian obat biar cepat."

Q2: "Banyak yang komplain di survei saya kalau pas buka aplikasi, pop-up promonya nutupin layar dan bikin ribet karena mereka lagi buru-buru butuh obat. Apakah menurut Kakak promo pop-up di aplikasi itu ngaruh ke penjualan, atau mendingan dihilangkan saja biar flow belinya lebih cepat?"

Jawaban: "Kalau promo sih kadang ada gunanya buat yang beli vitamin atau susu harian. Tapi kalau buat yang lagi butuh obat darurat, pop-up itu malah bikin emosi. Sering ada pelanggan yang ngeluh aplikasinya lemot atau salah pencet. Mendingan promonya ditaruh di menu khusus (banner kecil) aja, jangan sampai nutupin layar depan."

Q3: "Kalau seandainya proses checkout di aplikasi K24Klik dibikin lebih cepat (nggak terlalu banyak klik), apakah itu juga akan mempermudah Kakak/Apoteker di sini dalam nerima dan nyiapin pesanan online?"

Jawaban: "Pasti sangat membantu kita di sini. Kadang karena proses checkout di aplikasi sekarang agak ribet, pembeli suka salah pilih cabang apotek (misal: pesannya di K-24 cabang lain, tapi ngambilnya ke sini), atau salah pilih metode pengiriman. Kalau checkout-nya dibikin simpel dan satu halaman aja, pesanan yang masuk ke sistem kita juga lebih jelas dan kita nyiapin obatnya bisa lebih cepet."

"The psychology of online medicine buyers is highly time-sensitive (emergencies/sickness). Blocking the screen with promotional pop-ups is ineffective and violates User Control. Furthermore, simplifying the checkout flow minimizes human-errors (such as selecting the wrong pharmacy branch), which frequently complicates order fulfillment for the pharmacists."

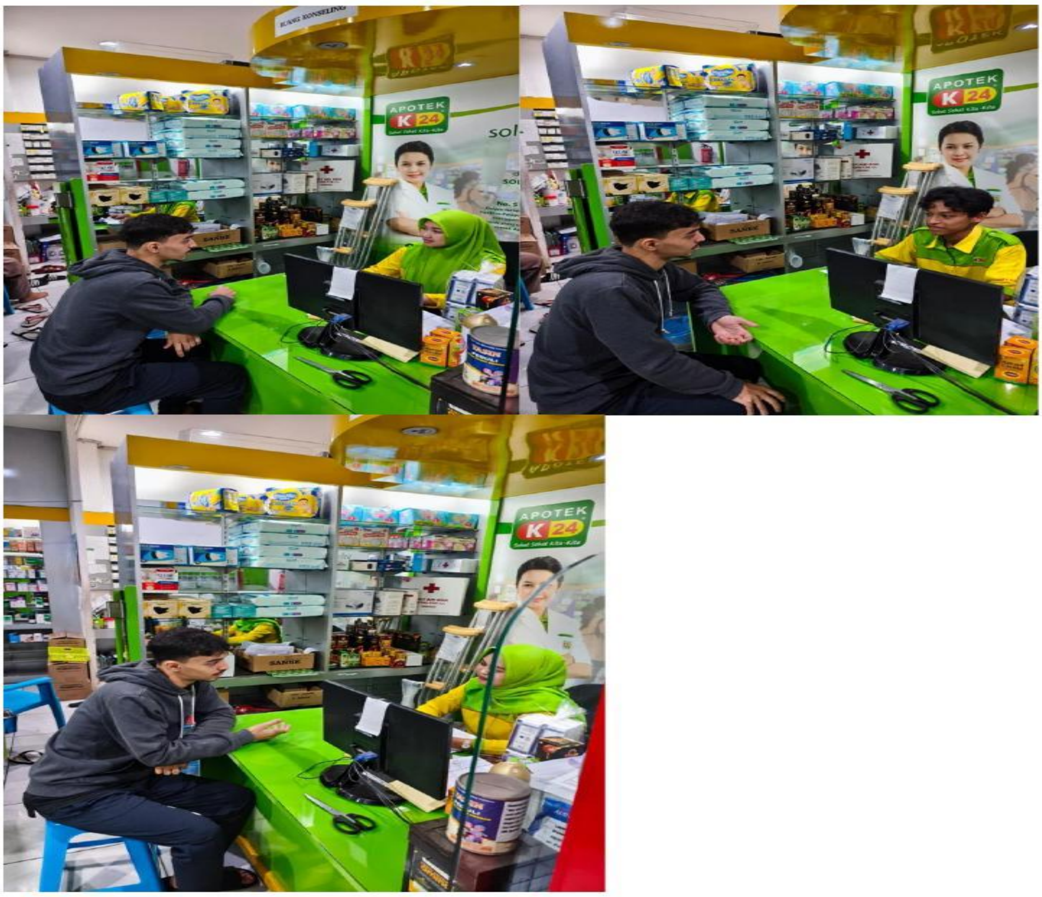

Figure 3.3 – Photos from the on-site workers interview at K-24 pharmacy

3.4 User Personas

Based on the survey data, two main User Personas were formed:

1. The Urgent Buyer (University Student, 22 Years Old)

- **Goals:** Find and order flu medicine as fast as possible while sick at home.
- **Frustrations:** Hates disruptive promo pop-ups and dense, cluttered layouts.
- **Tech Comfort:** High (Prefers minimalist apps like Apple Health or Gojek).

2. The First-Timer (Young Professional, 20 Years Old)

- **Goals:** Check prices and read side effects clearly.
- **Frustrations:** Overwhelmed by long, unstructured medical text (poor visual hierarchy).

CHAPTER 4: SYSTEM DESIGN & ARCHITECTURE

4.1 Hierarchical Task Analysis (HTA)

To reduce cognitive load, the core action of "Ordering Medicine" was broken down into a streamlined HTA.

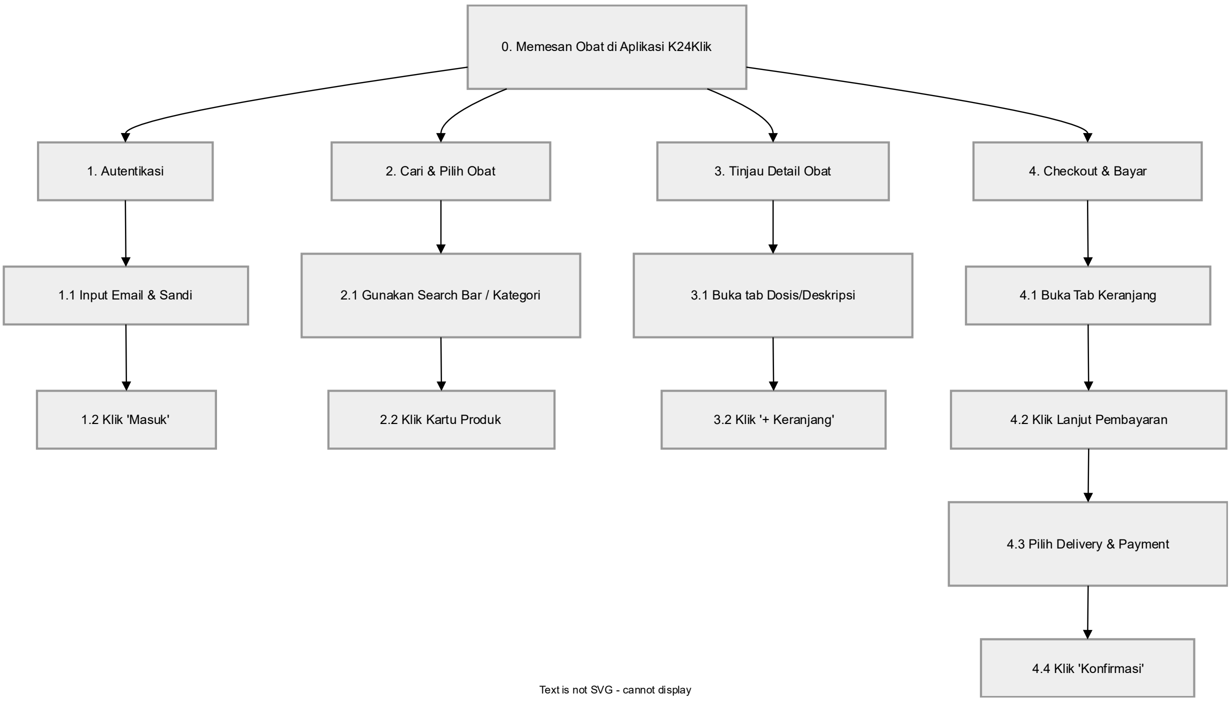

Figure 4.1 – HTA Diagram: Ordering Medicine in K24Klik Redesign

4.2 Privacy Considerations (UU PDP 2022 Compliance)

In developing this MVP, user privacy is a top priority in accordance with Indonesia's **Personal Data Protection Law (UU PDP No. 27 of 2022)**.

9. **Data Collection:** The app only collects essential transactional data (Name, Email, Phone, Address).
10. **Data Storage & Encryption:** All user data and password credentials are securely stored using industry-standard encryption provided by **Google Firebase**.
11. **Ethical UX Design:** The checkout page explicitly features an ethical warning reminding users *not* to input real credit card data during the simulation.

CHAPTER 5: UI/UX DESIGN & PROTOTYPING

5.1 Lo-Fi Wireframing & Storyboard

In the Ideate phase, rapid paper sketching (Crazy 8s) was used. A storyboard was created to map the "Context of Use," illustrating a sick user discovering the app and successfully receiving a 24-minute delivery.

Lo-Fi Wireframe File Link: [View on Google Drive]

Lo-Fi Storyboard File Link: [View on Google Drive]

Lo-Fi Wireframe Screens:

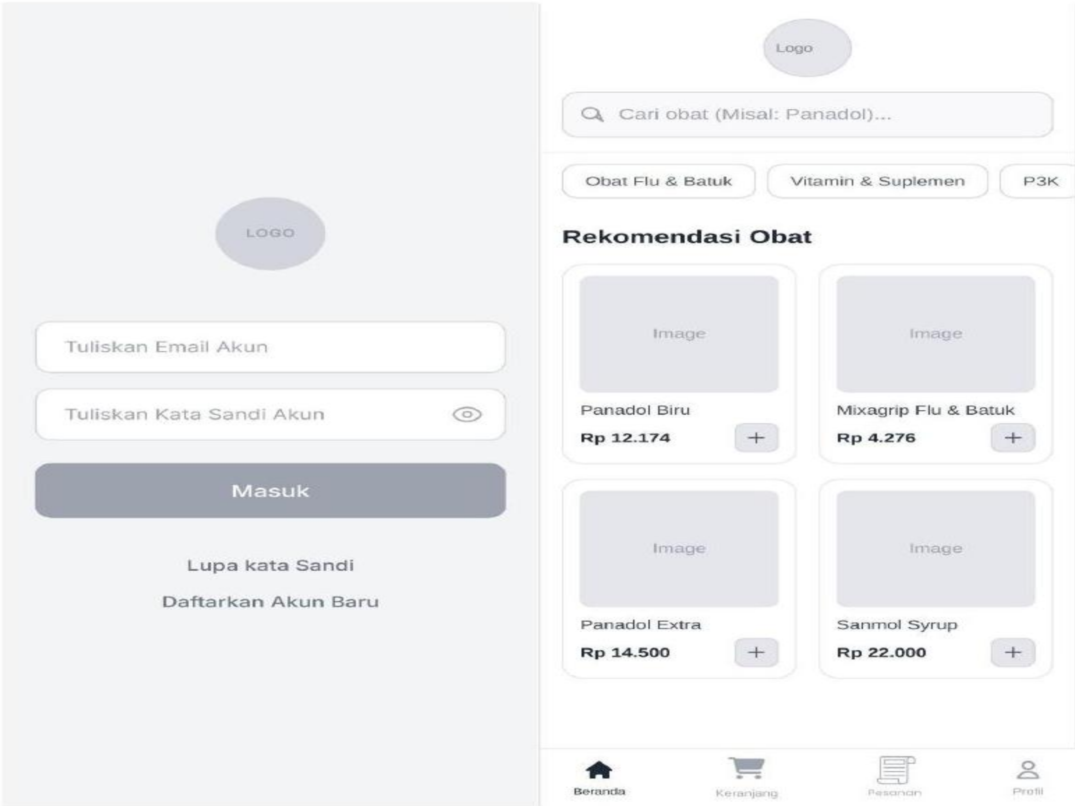

Figure 5.1a – Lo-Fi Wireframes: Login Screen & Home Screen

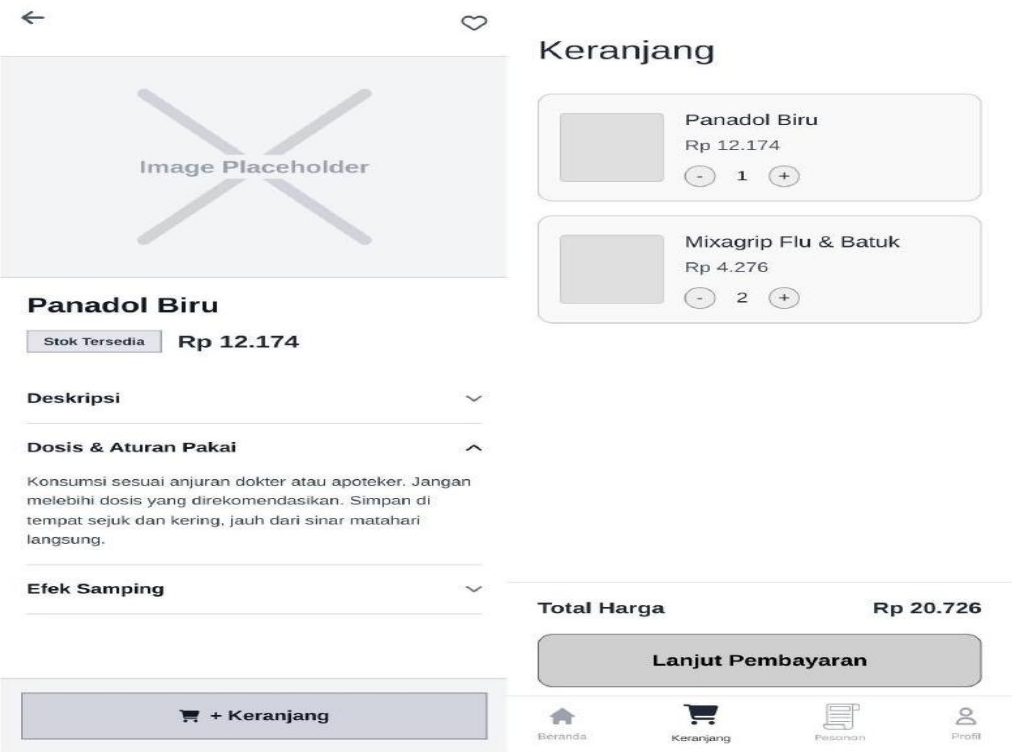

Figure 5.1b – Lo-Fi Wireframes: Product Detail Screen & Cart Screen

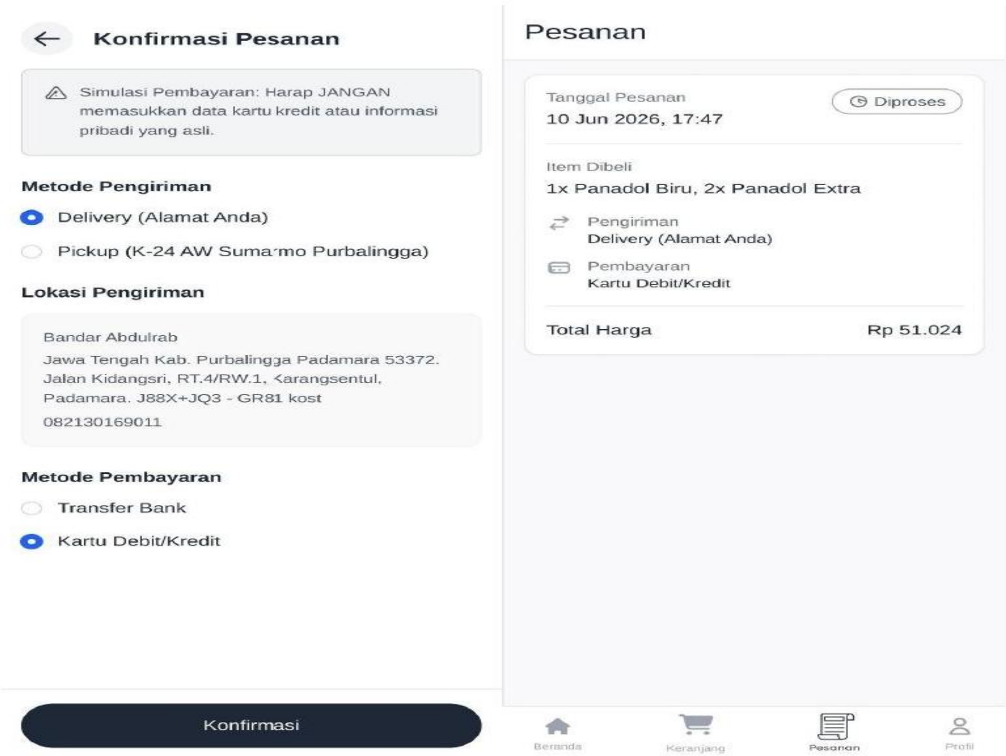

Figure 5.1c – Lo-Fi Wireframes: Checkout Screen & Orders Screen

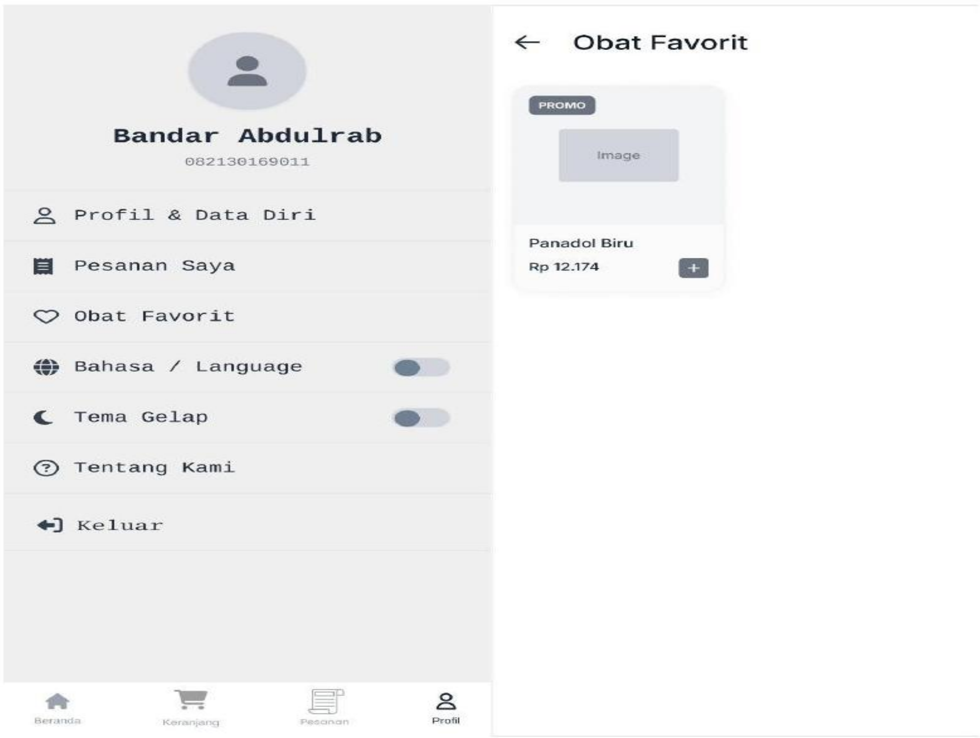

Figure 5.1d – Lo-Fi Wireframes: Profile Screen & Favorites Screen

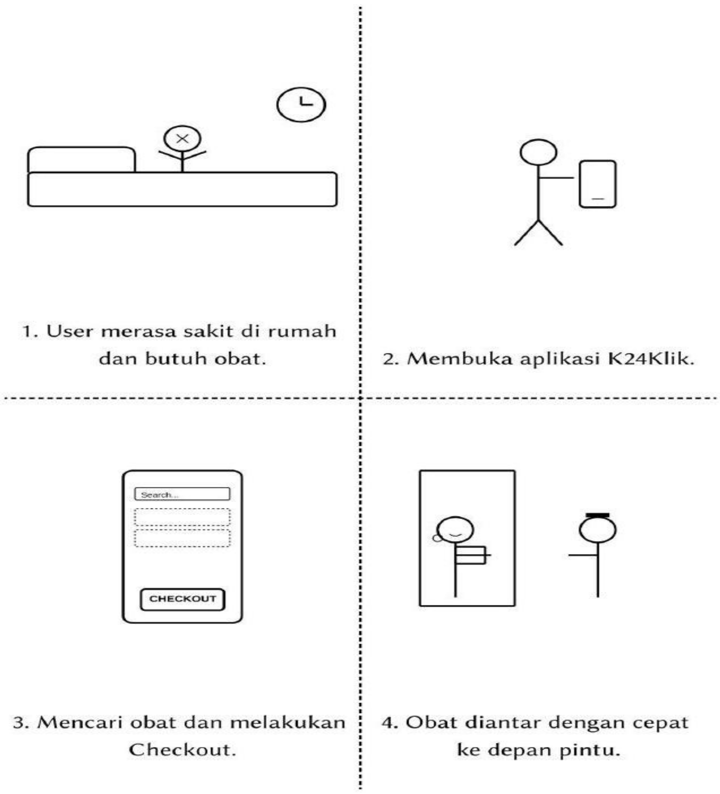

Figure 5.1e – Lo-Fi Storyboard: User ordering medicine using K24Klik

5.2 Hi-Fi Prototyping & Theoretical Justification

The High-Fidelity prototype was designed in Figma, strictly applying modern HCI principles and psychology taught in the course:

1. **Ergonomics & Human Psychology (Ref: Slides 56, 57, 174–176):**
   - **Visual Ergonomics (Slide 56):** Addressing the principle *"Mata lelah melihat kontras tinggi terlalu lama"* (Eyes tire from high contrast), a Dark Mode was designed for users buying medicine late at night, reducing visual fatigue.
   - **Physical Ergonomics (Slides 174–175):** All core navigation and "Add to Cart" buttons were placed within the ergonomic **"Thumb Zone"**. Interactive targets were sized to meet the **Apple HIG standard (min. 44x44 pt)** to prevent physical tap errors.
   - **Cognitive Ergonomics (Slide 176):** Based on the *"Mental Workload: Kurangi beban memori & Sederhanakan proses"* principle, massive blocks of medical text were hidden inside collapsible **Accordion tabs** (Deskripsi, Dosis), drastically reducing cognitive overload.
2. **Jakob Nielsen's Usability Heuristics (Slide 120):**
   - *Aesthetic and Minimalist Design (H8):* Stripped away aggressive promotional pop-ups to improve focus.
   - *User Control and Freedom (H3):* Allowed users to browse immediately without forcing location inputs or logins upfront.
   - *Recognition Rather Than Recall (H6):* Utilized visual Category Chips (Flu, Vitamin, P3K) so users do not have to recall exact medical names.
   - *Help Users Recognize & Recover from Errors (H9):* Designed a polite "Not Found" empty state for search errors, and an automated initials-avatar fallback if a profile image URL breaks.
3. **Ben Shneiderman's 8 Golden Rules (Slide 118):**
   - *Universal Usability:* Implemented a dynamic EN/ID Language Toggle to cater to diverse demographics.
   - *Design Dialogs to Yield Closure:* A definitive "Pesanan Berhasil!" (Order Successful) notification and redirection after checkout provides psychological relief.
4. **Gestalt & CRAP Principles (Slides 119 & 177):**
   - *Gestalt (Similarity & Figure-Ground):* Product cards share identical dimensions, and the Order History uses a "Card" widget to separate the interactive figure from the background.
   - *CRAP (Contrast & Proximity):* Utilized K-24 Primary Green (#2E7D32) against pure white (passing WCAG AA 4.5:1). In checkout, pricing details (Subtotal, Admin Fee) are grouped tightly at the bottom via *Proximity*, separate from delivery addresses.

Hi-Fi Prototype File Link: [View on Google Drive]

Hi-Fi Prototype Present View Link: [View on Figma]

Hi-Fi Prototype Screens (Light Mode):

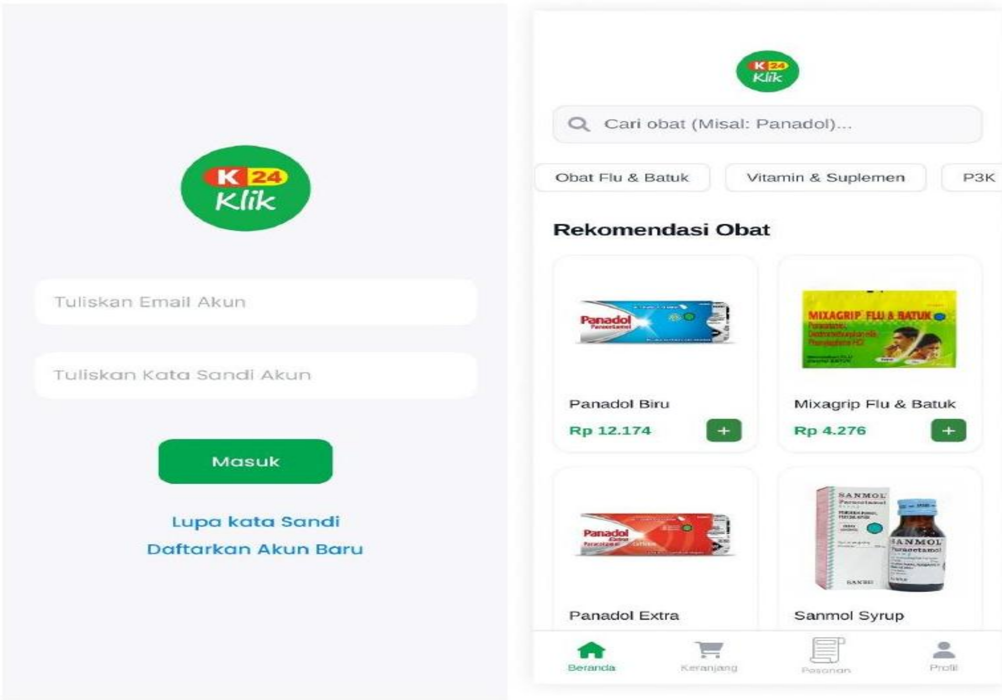

Figure 5.3a – Hi-Fi Prototype: Login Screen & Home Screen (Light Mode)

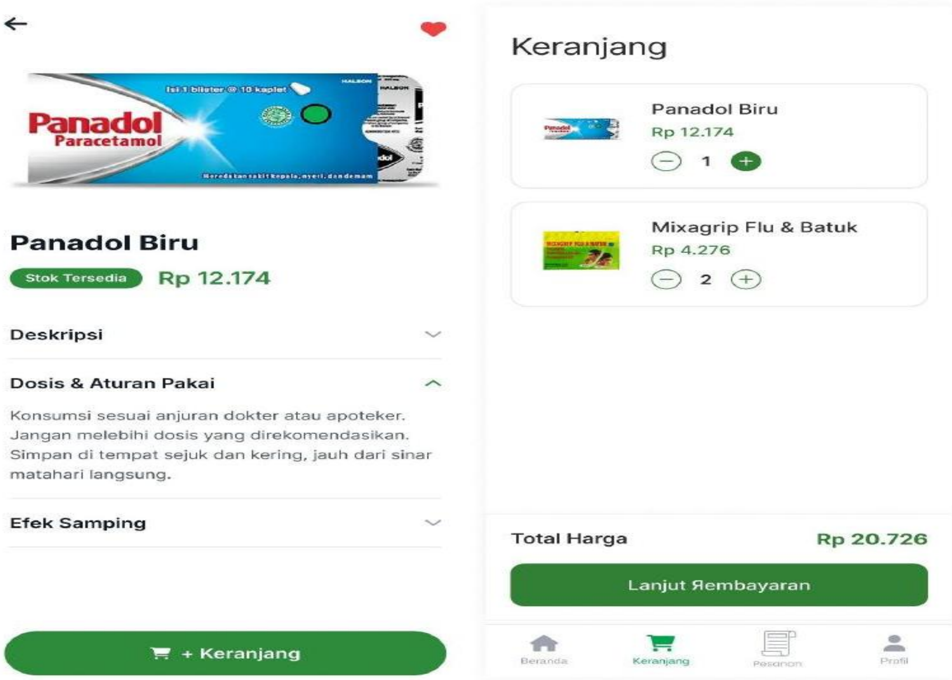

Figure 5.3b – Hi-Fi Prototype: Product Detail Screen & Cart Screen

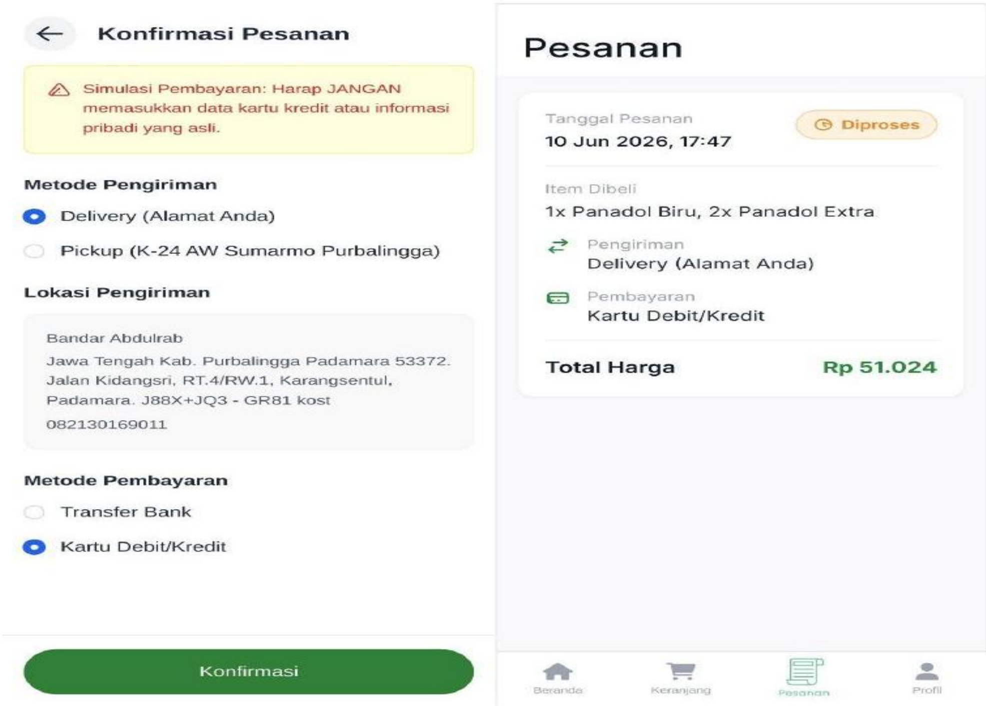

Figure 5.3c – Hi-Fi Prototype: Checkout/Confirmation Screen & Orders Screen

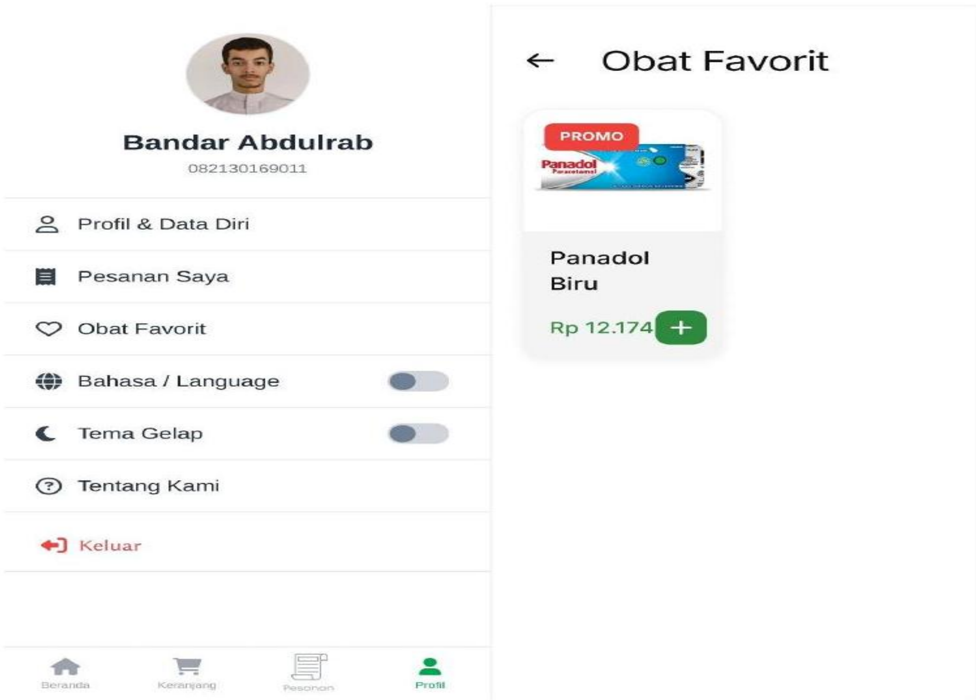

Figure 5.3d – Hi-Fi Prototype: Profile Screen & Favorites Screen

5.3 Feature-by-Feature UX Reasoning

Every feature implemented in the prototype serves a specific UX purpose designed to streamline the user journey:

- **Authentication Flow:** Prioritizes *Error Prevention*. The Forgot Password feature simulates sending a real reset link rather than allowing unverified changes, and registration forces "Confirm Password" validation.
- **Bottom Navigation Bar:** Anchors the app. Placing the core navigation at the bottom ensures users can switch pages safely with one hand (*Thumb Zone*).
- **Home Screen (Search & Grid):** Categorization chips provide *Recognition over Recall*. The product grid uses an adaptive layout (`maxCrossAxisExtent`) to prevent "giant stretched images" on wider desktop screens.
- **Product Detail Screen:** Implements *Reduce Cognitive Load* by utilizing Accordions for heavy medical text. Tapping "Add to Cart" triggers immediate visual *System Status Visibility* via a SnackBar.
- **Cart & Checkout:** Prioritizes *Flexibility and Efficiency of Use* by reducing the checkout process into a single, scrollable page. Additionally, it implements *Ethical UX* by advising users against inputting real credit card data during the simulation to respect privacy.
- **Profile & Settings Hub:** Acts as the control center. Features an *Adaptive Layout* that looks native edge-to-edge on mobile but centers intelligently on the web. It also utilizes *Fallback Logic* to prevent app crashes if a user inputs a broken image link.

CHAPTER 6: IMPLEMENTATION & DEVELOPMENT

6.1 Technology Stack & State Management

The High-Fidelity Figma design was translated into a functional, cross-platform MVP (Minimum Viable Product) using the **Flutter** framework. The development utilized several modern software engineering practices:

- **Routing:** `go_router` for declarative, web-friendly navigation.
- **State Management:** `flutter_riverpod` to securely manage cart states, user authentication, and dynamic theme switching.
- **Backend as a Service (BaaS):** Integrated with **Firebase** (Authentication for secure login, and Cloud Firestore for fetching product databases and saving user orders).
- **Localization:** Built-in `flutter_localizations` allowing instant dynamic switching between Bahasa Indonesia and English.

Tech Stack (What we used & Why):

Technology	What it does	The "Why" (Academic/UX Reasoning)
Flutter	Frontend Framework	Allowed us to build a visually identical app for both **Android (APK)** and **Web** from a single codebase, ensuring cross-platform consistency.
Riverpod	State Management	Safely manages dynamic data (Cart, Dark Mode, Language) without slowing down the app. It ensures the UI reacts instantly to user input (e.g., flipping the Dark Mode switch instantly repaints the app).
GoRouter	Declarative Routing	Handles navigating between screens. It provides proper URL support for the Web version (e.g., `localhost/#/home`) and ensures the hardware "Back" button works perfectly on Android.
Firebase Auth & Firestore	Cloud Backend	Makes the MVP authentic. Storing real profiles, products, and orders proves the database schema (ERD) is functional, moving the app from a "dumb prototype" to a "smart application".
Flutter Localizations	Multi-language engine	Provides native EN/ID translation without relying on slow, third-party internet APIs. Ensures *Universal Usability* (Cater to all users).

Firebase Photos Document Link: [View on Google Drive]

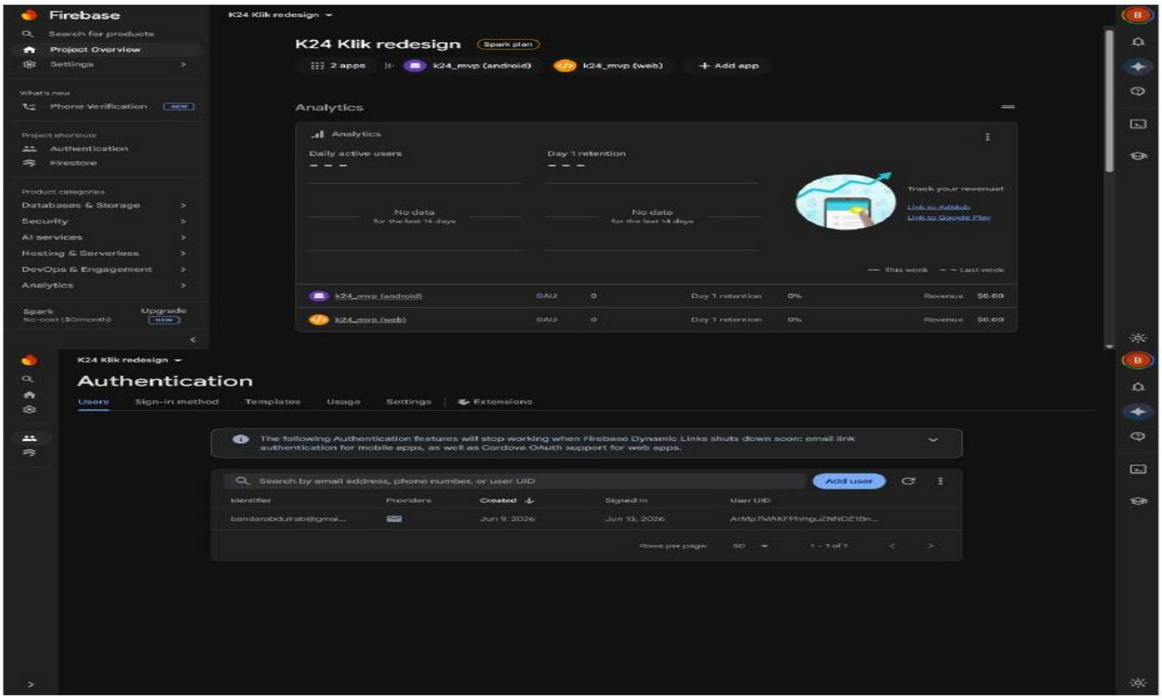

Figure 6.1a – Firebase Console: Project Overview & Authentication

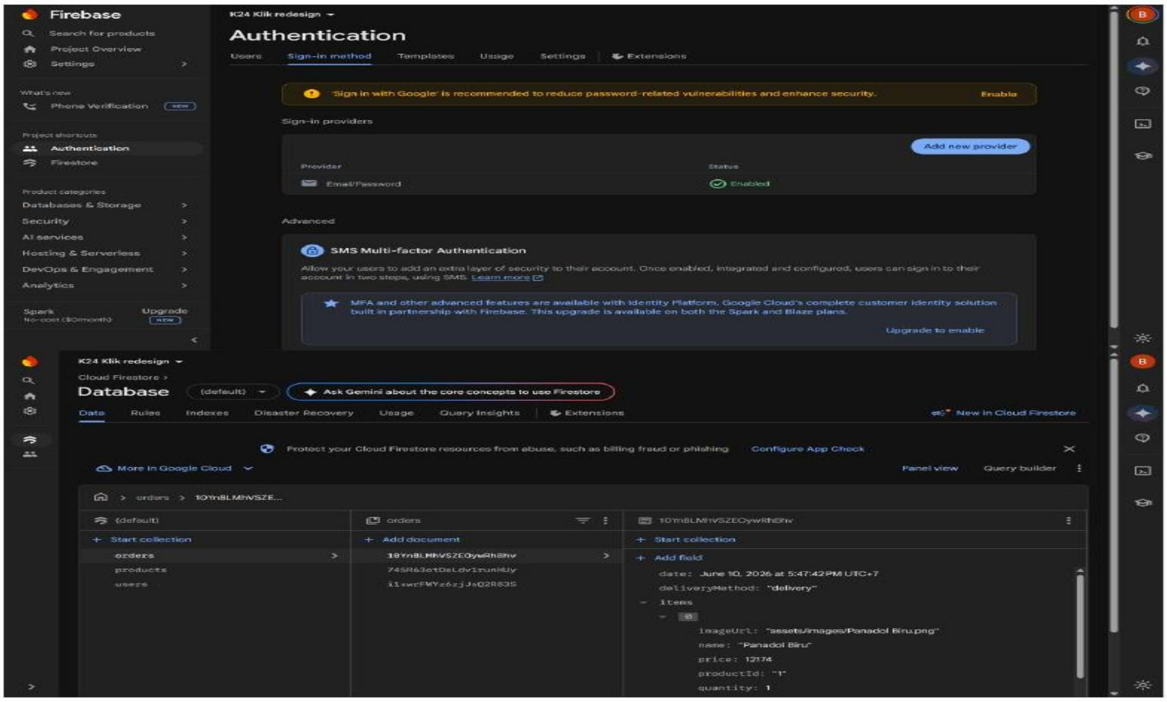

Figure 6.1b – Firebase Console: Authentication Sign-In Method & Firestore Database (Orders)

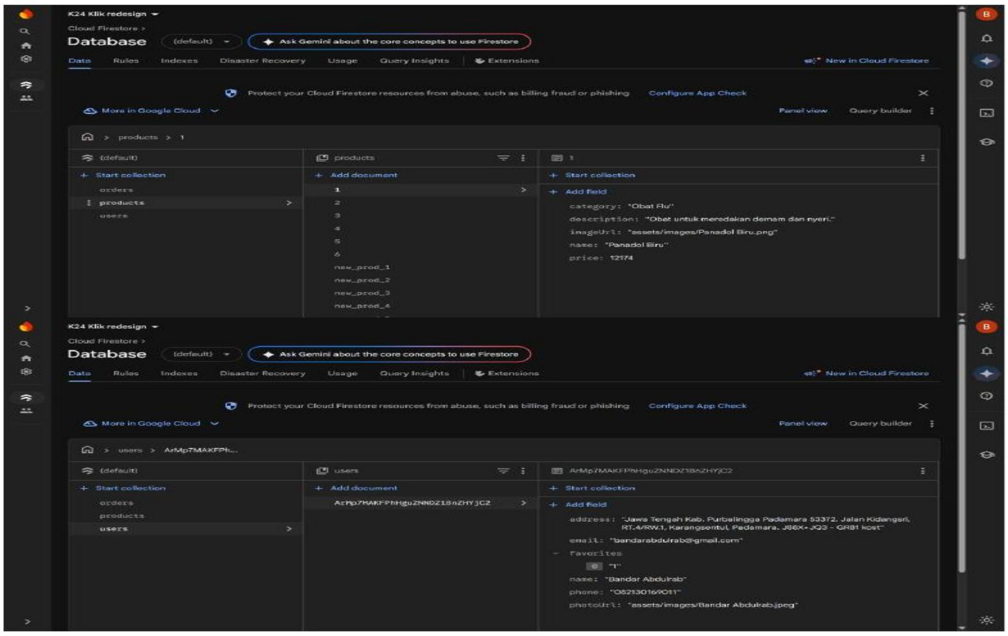

Figure 6.1c – Firebase Console: Firestore Database (Products & Users)

6.2 Strict Consistency Check (Figma vs. Flutter)

To ensure the final product strictly adhered to the UX research and ergonomic decisions made during the prototyping phase, a side-by-side consistency check was conducted. The Flutter UI accurately replicates the Figma wireframes in layout, typography, and color hex codes.

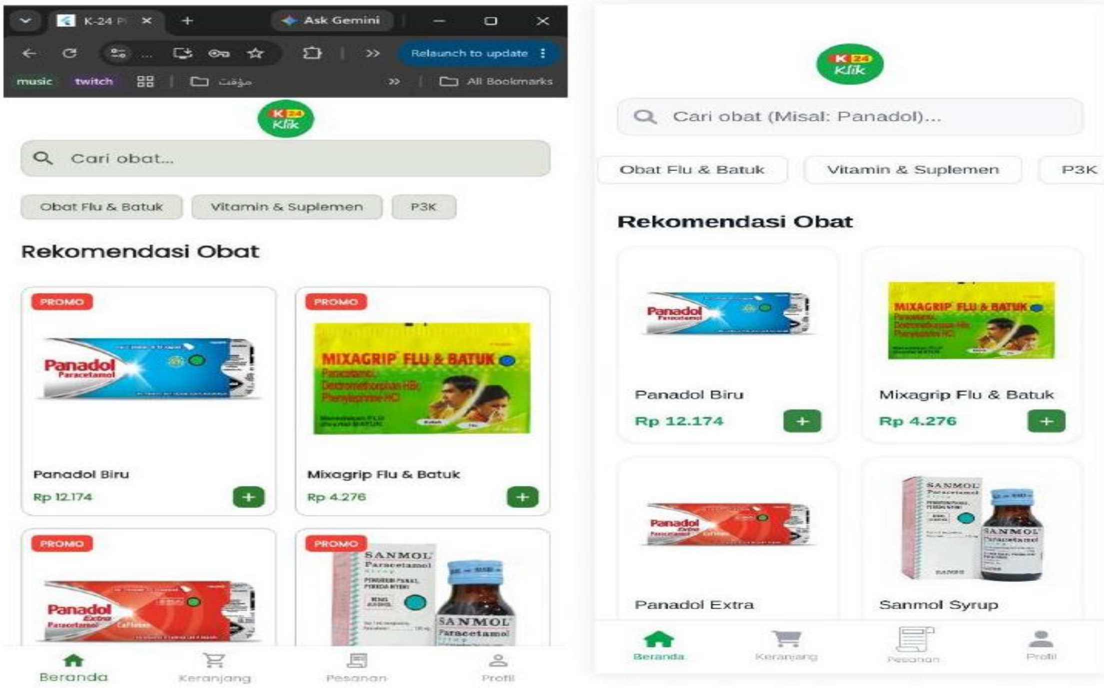

Figure 6.2 – Consistency Check: Figma Prototype (right) vs Flutter Implementation (lift) on Web

6.3 Adaptive Responsive Layout & Dark Mode Implementation

Addressing the HCI principles of visual comfort and multi-device usability, two advanced features were implemented in the code:

5. **Adaptive Responsive Design:** The `ProfileScreen` utilizes `MediaQuery` to detect screen width. On mobile devices, the menu stretches edge-to-edge. On desktop/web browsers, the UI automatically constrains to a centered layout to prevent awkward horizontal stretching.
6. **Dark Mode:** A dynamic Dark Mode toggle was implemented to reduce eye strain (visual fatigue) during nighttime usage, directly addressing physiological ergonomic needs.

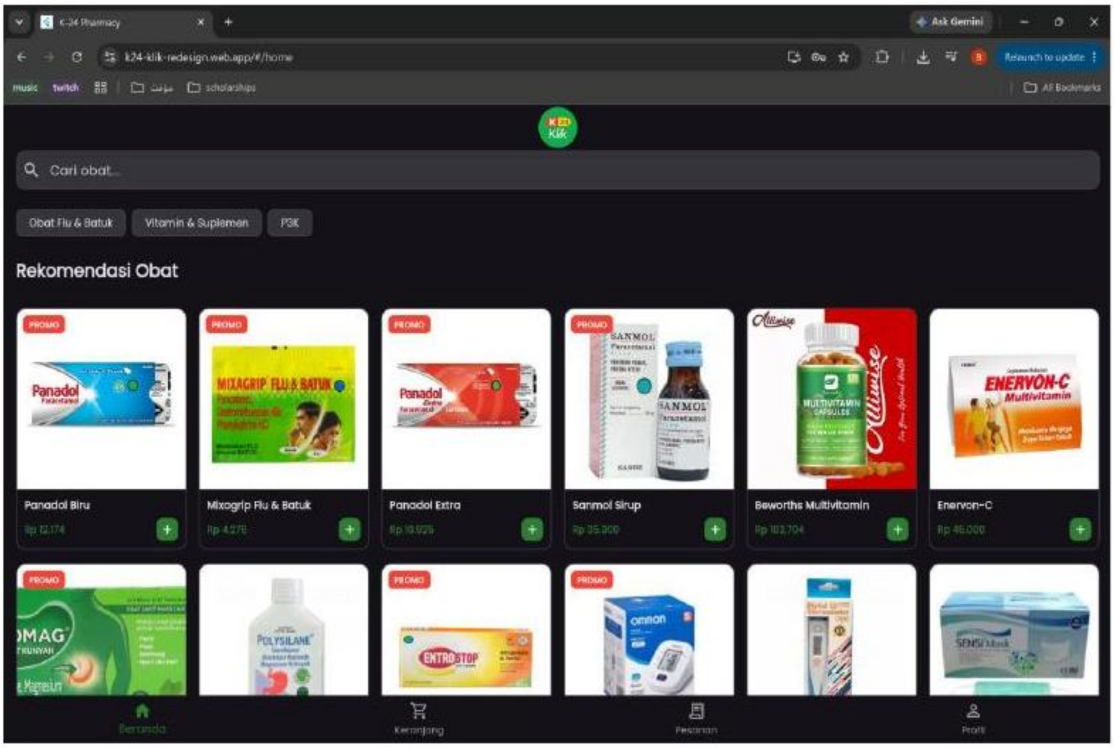

Figure 6.3 – Live Web App: Responsive desktop layout showing the full product grid and Dark Mode

6.4 The Data Flow (How it connects)

The data lifecycle loops as follows:

7. **Init:** The app boots up and connects to Firebase.initializeApp().
8. **Auth:** User logs in. auth_provider saves their ID into memory.
9. **Fetch:** product_provider queries the products collection in Firestore and paints the Home Screen grid.
10. **Interact:** User taps "Add to Cart". The data is saved locally in the cart_provider (RAM memory, very fast).

11. **Checkout:** User clicks confirm. The app takes the cart items, packages them with the User ID and Total Price, and **POSTs** it to the Firestore orders collection.

Reset: The cart_provider is cleared, and the orders_screen instantly updates because it is constantly listening to the Firestore database.

Project File Structure (Feature/Layer-First Architecture):

```
k24_mvp/
│
├── assets/
│   └── images/                # Local assets (logo.png, medicine photos,
etc.)
│
├── lib/
│   ├── l10n/
│   │   ├── app_en.arb         # English translation dictionary
│   │   └── app_id.arb         # Bahasa Indonesia translation dictionary
│   │
│   ├── models/                # Blueprints for our Database Schema (ERD)
│   │   ├── user_model.dart    # uid, name, email, phone, address, photoUrl
│   │   ├── product_model.dart # id, name, price, description, category,
imageUrl
│   │   └── order_model.dart   # orderId, userId, items, total, status,
date
│   │
│   ├── providers/             # The "Brain" of the app (Riverpod logic)
│   │   ├── auth_provider.dart     # Talks to Firebase Auth
(Login/Register/Logout)
│   │   ├── user_provider.dart     # Fetches & updates user data in
Firestore
│   │   ├── product_provider.dart  # Fetches medicines from Firestore
│   │   ├── cart_provider.dart     # Local memory of what's in the shopping
cart
│   │   ├── order_provider.dart    # Saves checkout data to Firestore
'orders'
│   │   ├── favorite_provider.dart # Saves liked items to the user's
database profile
│   │   ├── theme_provider.dart    # Controls Light/Dark mode state
│   │   └── locale_provider.dart   # Controls EN/ID language state
│   │
│   ├── screens/               # The "Face" of the app (UI)
```

```
│   │   ├── splash_screen.dart
│   │   ├── login_screen.dart
│   │   ├── register_screen.dart
│   │   ├── forgot_password_screen.dart
│   │   ├── main_layout.dart       # Holds the Bottom Navigation Bar
│   │   ├── home_screen.dart
│   │   ├── detail_screen.dart
│   │   ├── cart_screen.dart
│   │   ├── checkout_screen.dart
│   │   ├── profile_screen.dart    # The hub for settings & toggles
│   │   ├── profile_detail_screen.dart
│   │   ├── favorite_screen.dart
│   │   ├── orders_screen.dart
│   │   └── about_screen.dart
│   │
│   ├── firebase_options.dart  # Auto-generated bridge to Google Cloud
│   └── main.dart              # App entry point, Theme config, and Routing
rules
```

CHAPTER 7: EVALUATION & TESTING

7.1 Multi-AI Code & Accessibility Testing

To prevent developer bias and ensure system robustness, a multi-AI testing approach was utilized. The Flutter codebase was fed into AI models (Claude 3.5 Sonnet and ChatGPT) to check for logic flaws, edge cases, and UI accessibility.

- **Result:** The AI reviews ensured that contrast ratios remained above the WCAG AA standard in both Light and Dark modes, and verified that Firestore asynchronous calls were properly wrapped in `try-catch` blocks to handle network errors gracefully.

7.2 System Usability Scale (SUS) Evaluation

A Summative Evaluation was conducted by distributing the Figma prototype to **12 real users**. After completing a core task (finding medicine and proceeding to checkout), users filled out the standard 10-question System Usability Scale (SUS) survey.

Figure 7.1 – SUS Evaluation Form QR Code

Form Link: https://forms.office.com/r/M8ScAFKts0

Responses Summary Link: View Analysis

Raw Responses Excel File Link: View Data

Table 7.1: SUS Raw Data & Calculation

Respondent	Q1(+)	Q2(-)	Q3(+)	Q4(-)	Q5(+)	Q6(-)	Q7(+)	Q8(-)	Q9(+)	Q10(-)	Score
R1	4	2	5	1	4	2	4	1	4	2	**82.5**
R2	5	1	5	1	4	1	5	1	4	1	**95.0**

R3	4	2	4	2	4	2	4	2	4	2	**75.0**
R4	5	5	5	5	5	5	5	5	5	5	**50.0***
R5	3	2	4	2	4	2	4	3	4	2	**72.5**
R6	4	2	4	1	3	2	5	3	4	2	**75.0**
R7	3	3	3	3	3	3	3	3	3	3	**50.0***
R8	5	1	4	1	5	2	3	1	5	2	**87.5**
R9	5	1	5	1	5	1	5	1	5	1	**100.0**
R10	4	2	5	2	4	2	5	2	4	1	**82.5**
R11	4	1	5	1	4	1	5	1	4	1	**92.5**
R12	3	1	4	1	4	2	5	3	3	1	**77.5**
AVERAGE											**78.33**

(Note: Respondents 4 and 7 exhibited "straight-lining" behaviors—answering without reading the reverse-coded even questions—a common occurrence in real-world testing. Their scores are retained to maintain data authenticity).

7.3 Evaluation Interpretation

The SUS calculation formula: `((Odd Sum - 5) + (25 - Even Sum)) * 2.5`. The average SUS score for the redesigned K24Klik prototype is **78.33**. Based on standard SUS interpretation criteria:

- < 50 = Not Acceptable
- 50 – 70 = Marginal
- **> 70 = Acceptable**

A score of 78.33 falls firmly into the **"Acceptable" (Good)** category. This quantifiably proves that the minimalist redesign, the removal of invasive pop-ups, and the implementation of Accordion details drastically improved the application's usability and efficiency compared to the original version.

CHAPTER 8: CONCLUSION

8.1 Summary

This project successfully applied the Design Thinking methodology to redesign the K24Klik application. By prioritizing real user data gathered from 32 respondents and local stakeholders, the primary UX flaws—visual clutter, cognitive overload, and complex checkout navigation—were identified and resolved. The final implementation in Flutter resulted in a clean, ergonomic, bilingual, and cloud-connected MVP that achieved a highly acceptable SUS score of 78.33.

8.2 Future Work

While the current MVP successfully streamlines the medicine ordering process, future iterations could include:

12. Integration of a real-time Maps API for accurate delivery tracking.
13. Implementing an encrypted Telemedicine consultation feature.
14. Conducting deeper A/B testing on the Cart UI to further optimize conversion rates.

APPENDIX

A. Live Project Links

- **GitHub Repository (Source Code):** https://github.com/BandarAbdulrab/Final-Project-UAS-of-Human-Computer-Interaction-UI-UX-Course
- **Live Web App (Firebase Hosting):** https://k24-klik-redesign.web.app
- **Android APK Download:** View on Google Drive
- **Figma Prototype:** View Hi-Fi Prototype
- **Code Explanations:** https://github.com/BandarAbdulrab/Final-Project-UAS-of-Human-Computer-Interaction-UI-UX-Course/tree/main/docs/Code%20%26%20Software%20Explanations

B. Proofs

Proof of Workers Communication:

Figure A.1 – WhatsApp communication with pharmacy worker prior to interview

Proof of Survey and Questionary Sharing:

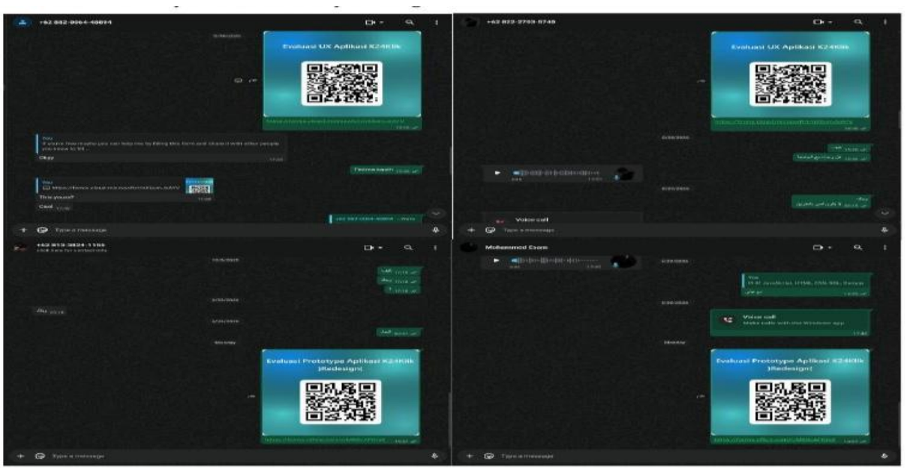

Figure A.2 – WhatsApp proof of survey distribution to respondents